

Graph Theory – Lecture Notes

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1 Introduction

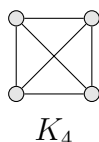
Definition 1.1. A (*simple*) *graph* G is a pair (V, E) where V is a non-empty set and $E \subseteq \{\{x, y\} \in V \times V : x \neq y\}$. We call the elements of V *vertices* of G and the elements of E *edges* of G . We also write $V(G)$ for V and $E(G)$ for E , respectively.

‘Simple’ means no loops and no double edges.

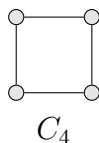
Remark 1.2. V is finite, unless otherwise stated.

Example 1.3. We define a few common graphs. Let $n, m \in \mathbb{N}$.

- The *complete graph* K_n has n vertices and each pair of vertices is connected.



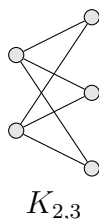
- The *cycle* C_n has n vertices, which are cyclically connected.



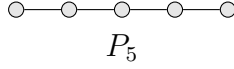
- Let A and B be disjoint sets with $|A| = n$ and $|B| = m$. The *complete bipartite graph* $K_{m,n}$ is defined by

$$V = A \dot{\cup} B \tag{1}$$

$$E = \{\{a, b\} : a \in A, b \in B\} \tag{2}$$



- The *path* P_n has the vertices $1, \dots, n$ and edges $\{a, a+1\}$ for $a \in \{1, \dots, n-1\}$. (in books sometimes P_{n-1} has n vertices)



Definition 1.4. Let G be a graph, $u, v \in V(G)$

- If $\{u, v\} \in E(G)$, then u and v are **adjacent** and the edge $e = \{u, v\}$ is **incident** with both u and v .
- $N_G(v) = \{w \in V(G) : \{v, w\} \in E(G)\}$ is the **neighbourhood** of v in G .
- $\deg_G(v) = |N_G(v)|$ is the **degree** of v .

Lemma 1.5 (Handshake Lemma). *Let G be a graph. Then*

$$\sum_{v \in V(G)} \deg_G(v) = 2|E(G)|. \quad (3)$$

(in particular the sum of the degree is even).

Proof. $\sum_{v \in V(G)} \deg_G(v) = \#\{(v, e) : v \in V(G), e \in E(G) \text{ such that } v \text{ incident to } e\} = 2|E(G)|$ $\square$

Definition 1.6. Two graphs $G = (V, E)$ and $G' = (V', E')$ are **isomorphic** if there exists an isomorphism ϕ between them, i.e. a bijection $\phi : V \rightarrow V'$ such that $\forall u, v \in V(G) : \{u, v\} \in E(G) \iff \{\phi(u), \phi(v)\} \in E(G')$

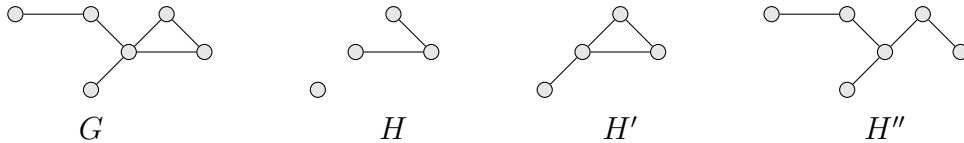
Definition 1.7. Let $G = (V, E)$ be a graph. A **subgraph** of G is a graph $H = (V', E')$ with $V' \subseteq V$ and $E' \subseteq E \cap \binom{V'}{2}$ where $\binom{S}{K}$ is the set of all K -subsets of S .

H' is an **induced subgraph** of G if it is a subgraph and $E' = E \cap \binom{V'}{2}$

H'' is a **spanning subgraph** if it is a subgraph and $V' = V$

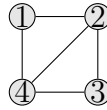
For $W \subseteq V$, write $G[W]$ for the subgraph induced by the vertices in W

Example 1.8.



Definition 1.9. Let v_i be vertices of a graph G

- A **walk** is a sequence of vertices v_i s.t. $\{v_i, v_{i+1}\} \in E(G) \forall i \in \{1, \dots, k-1\}$
- The **length** of a walk is k (number of vertices in the walk)
- The walk is **closed** if $v_1 = v_k$
- A **trail** is a walk that does not use an edge twice
- A **path** is a walk for which $v_1, \dots, v_k$ are pairwise distinct



$(1, 2, 3, 4, 2, 3)$ is a walk, but no trail

$(1, 2, 3, 4, 2)$ is a trail, no path

$(1, 2, 3, 4)$ is a path

Definition 1.10. Let G be a graph

- A graph G is **connected** if there is a walk between any pair of its vertices
- A **(connected) component** of G is an inclusion wise maximal connected subgraph

1.1 Hamiltonian Cycles

Definition 1.11. A **Hamiltonian cycle** in a graph G is a spanning subgraph that is a cycle (isomorphic to C_n for some $n \in \mathbb{N}$)

If G has a Hamiltonian cycle, G is called **Hamiltonian**

Theorem 1.12 (Dirac's Theorem). *Let G be a graph of order $n \geq 3$ where the **order** of G is given by $|V(G)|$ and the **size** by $|E(G)|$. If every Vertex of G has degree at least $\frac{n}{2}$ then G is Hamiltonian*

Theorem 1.13 (Ore's theorem). *Let G be a graph of order $n \geq 3$. If for every pair $u, v \in V, u \neq v, u$ and v non-adjacent, we have $\deg(u) + \deg(v) \geq n$ then G is Hamiltonian*

Remark 1.14.

Ore $\implies$ Dirac

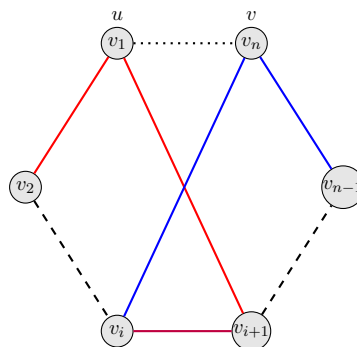
Proof.

The Graph G is connected: suppose G is not connected. Take an arbitrary component of G and call it H . The order of H is $n - k$ if k vertices are not in $V(H)$. Now any $v \in V(H)$ has $\deg(v) \leq n - k - 1$ with equality if v is adjacent to every vertex in $V(H)$. Now take $u \notin V(H)$ so $\deg(u) \leq k - 1$. It follows that $u, v \in V(G), u \neq v$ and u, v are not adjacent so $\deg(u) + \deg(v) \geq n$ but $\deg(u) + \deg(v) \leq k - 1 + n - k - 1 = n - 2 < n$ $\nmid$

- G has a Hamiltonian cycle: By induction on the number of non-edges
- No non-edge $\implies G$ complete, hence Hamiltonian
- Assume G non complete $u, v \in V, u \neq v$, be non-adjacent. By induction the graph G' obtained from G by adding $e = \{u, v\}$ as an edge is Hamiltonian. Let C be a Hamiltonian cycle of G' . If C does not use $e \implies C$ is Hamiltonian cycle in G . Assume C uses e . Removing e from C yields a path from u to v

For $i \in \{2, \dots, n - 1\}$

- If v_i is a neighbour of u , color $\{v_{i-1}, v_i\}$ red
- If v_i is a neighbour of v , color $\{v_i, v_{i+1}\}$ blue
- Since $\deg(u) + \deg(v) \geq n$ and the path $v_1, \dots, v_n$ has $n - 1$ edges, there exists an edge $\{v_i, v_{i+1}\}$ which is both red and blue. Then $v_1, \dots, v_i, v_n, v_{n-1}, \dots, v_{i+1}$ is a Hamiltonian cycle in G .



□

1.2 Eulerian Trail

Definition 1.15. An **Eulerian trail** is a trail that passes through each edge of the graph exactly once. A graph with a closed Eulerian trail is called **Eulerian**

Theorem 1.16 (Euler's theorem). *Let G be a connected graph with at least 2 vertices. Then G has a closed Eulerian trail iff every vertex in G has even degree*

Proof.

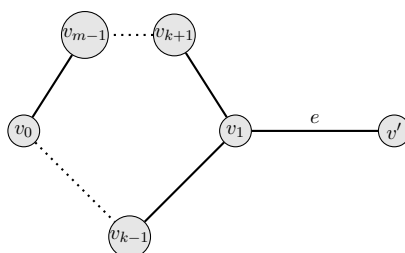
" $\implies$ "

Let T be a closed Eulerian trail and fix an orientation on T . Then for every $v \in V(G)$, the incident edges can be classified as "ingoing" or "outgoing". This defines a bijection between in and outgoing edges of T , so $\deg(v)$ is even.

" $\impliedby$ "

Let $T = (v_0, e_1, v_1, \dots, e_m, v_m)$ be a longest trail. We will prove (i) $v_0 = v_m$ and (ii) $\{e_i : i \in \{1, \dots, m\}\} = E(G)$. (i)+(ii) $\iff$ T is closed Eulerian trail.

- (i) If $v_0 \neq v_m$, then v_0 is incident to an odd number of edges in T . As $\deg(v_0)$ is even, there is $e \in E(G)$ incident to v_0 and not on T . We could thus append e on T $\nmid$ because T is longest trail.
- (ii) Claim: $V(T) = V$. Assume otherwise there exists a vertex $v' \notin V(T)$ and $e = \{v_k, v'\} \in E(G)$ for some $k \in \{0, \dots, m\}$ (by connectedness). Then $(v', e, v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, \dots, v_{k-1}, e_k)$ is a longer trail hence $V(T) = V$



Claim: $E(T) = E$. Assume otherwise. Consider $e \in E \setminus E(T)$ and write $e = \{v_k, v_l\}$. Then $(v_k, e_{k+1}, v_{k+1}, \dots, v_{m-1}, e_m, v_0, e_1, v_1, \dots, e_k, v_k, e, v_l)$ is a longer trail $\nmid$

□

Theorem 1.17. *Let G be a connected graph with at least two vertices. Then G has an Eulerian trail iff there are precisely 0 or two vertices of odd degree*

Proof.

By the Euler's Theorem, we know that if there are 0 vertices of odd degree then there exists an Eulerian trail (in fact a closed Eulerian trail), and if there exists a closed Eulerian trail, then all vertices have even degree.

Suppose now that G has exactly two vertices x and y of odd degree. We consider the following graph G' : Set $V(G') = V(G) \cup \{z\}$, where $z \notin V(G)$. Furthermore set $E(G') = E(G) \cup \{\{x, z\}, \{y, z\}\}$. Clearly, every vertex has even degree in G' , so there exists a closed Eulerian trail. Since z is only adjacent to x and y , these are the neighbours of z on the trail. Deleting the edges $\{x, z\}$ and $\{y, z\}$ from the trail yields an Eulerian trail from x to y in G . Conversely, if there is a non-closed Eulerian trail with endpoints x and y in G , then x and y are non-adjacent. Adding the edge $\{x, y\}$ creates a graph with a closed Eulerian trail, so all vertices have even degree by Euler's theorem. Thus x and y are the only vertices in G with odd degree. $\square$

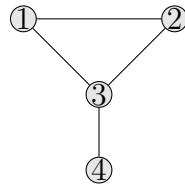
1.3 Datastructures

Assume that $G = (V, E)$ is a graph with $v = \{1, \dots, n\}$

Definition 1.18. The **adjacency matrix** of G is the matrix $A = (a_{ij})$ with
$$a_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & \text{else} \end{cases}$$

The **adjacency list** of G stores a label for each vertex, and a list of (pointers to) its neighbours.

Example 1.19.



$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

- $v_1: v_2, v_4$

- v_2 : v_1, v_3, v_4
- v_3 : v_2
- v_4 : v_1, v_2

Remark 1.20.

Memory requirement:

Adjacency matrix: $\theta(n^2)$

Adjacency list: $\theta(n + m)$, where m is the number of edges

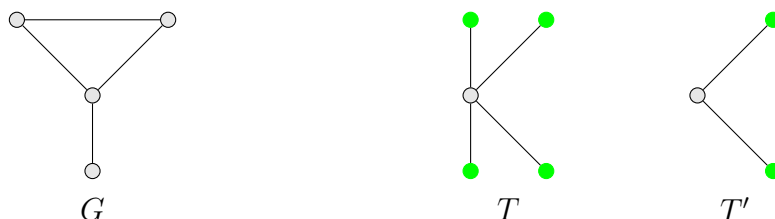
2 Trees

Definition 2.1. A **tree** is a graph that is connected and does not contain a cycle (a graph "contains a cycle" if it has a subgraph isomorphic to C_n for some $n \geq 3$).

A **forest** is a graph that does not contain a cycle.

A **leaf** is a vertex of a tree with degree 1.

Example 2.2.



- G is not a tree nor a forest
- T, T' are trees and forests
- $T \cup T'$ is a forest but not a tree
- All green vertices are leafs, all white vertices are not leafs

Lemma 2.3. *Every tree T with $n \geq 2$ vertices has at least one leaf.*

Proof. We iteratively construct a path $v_1, \dots, v_n$ as follows:

Start with any edge $\{v_1, v_2\} \in E(T)$. Assume that we have constructed $v_1, \dots, v_k$. To construct v_{k+1} , we may assume that v_k is not a leaf in T (or we are done). Otherwise v_k has a neighbour $v_{k+1} \neq v_{k-1}$. $v_{k+1} \neq v_i$ because T is a tree and otherwise we would get a cycle $(v_i, \dots, v_k, v_{k+1} = v_k)$. Add v_{k+1} to the path and continue finite many steps (because there are only finitely many vertices). The algorithm just stated ends at a leaf. $\square$

Theorem 2.4. *Let G be a graph with n vertices. The following are equal:*

- G is a tree
- G is connected with $n - 1$ edges
- G is connected and removing an edge disconnects it
- G has exactly one path between each pair of distinct vertices

(v) G has no cycle, but adding an edge creates a cycle

Proof. "1. $\implies$ 2."

Induction on n :

$n = 1$ is trivial

Let $n \geq 2$. By Lemma (2.2), G has a leaf v . Removing v and the unique edge incident to v yields a tree G' . By induction G' has $n - 2$ edges so G has $n - 1$ edges.

"2. $\implies$ 3."

Suppose there is an edge $e \in E(G)$ and the resulting graph by removing e is still connected. Keep removing as long as the resulting graph is still connected. Let G' be the resulting graph. Observe that G' is a spanning subgraph of G . G' does not contain a cycle as otherwise we could remove one edge from the cycle. By "1. $\implies$ 2.", applied to G' , we know that G' has $n - 1$ edges $\nless$ because G had $n - 1$ edges and we removed at least one.

"3. $\implies$ 4."

Suppose that there is a pair with $u, v \in V(G)$ s.t. there are two distinct paths u_i, v_i between them. Let i_0 be the maximum index s.t. $u_i = v_i : \forall i \in \{1, \dots, i_0\}$. The edge $\{v_{i_0}, v_{i_0+1}\}$ does not lie on the path u_i . The graph obtained by removing $\{v_{i_0}, v_{i_0+1}\}$ is connected as for any walk using this edge we can use u_i to go to u to reach v_{i_0} or v to reach v_{i_0+1} .

"4. $\implies$ 5."

If G had a cycle, then there would exist two paths between each pair of neighbouring vertices on the cycle (clockwise + counter clockwise).

Adding an edge creates a cycle: Let $u, v \in V(G)$ non adjacent. There is one path between the two. Now adding an edge between u, v creates a second path $\nless$

"5. $\implies$ 1."

We only need to show that G is connected. Suppose u, v are not connected by a walk. Creating an edge between u, v does not result in a cycle because for being isomorphic to C_n one needs at least two walks between each of the vertices in the cycle but by assumption the only walk connecting u, v is the newly added edge $\nless$ $\square$

Corollary 2.5. *Every connected graph G has a spanning tree.*

Proof.

Consider graphs on n vertices, now do induction on the number of edges m . As G

is connected, $m \geq n - 1$. If $m = n - 1 \implies G$ is a tree. If $m > n - 1 \implies G$ is not a tree. By (2.3) we can remove edges s.t. the resulting graph is connected. By induction it has a spanning tree, which is then also a spanning tree of G . $\square$

Theorem 2.6 (Matrix-tree theorem). *Let G be a graph with vertices $\{1, \dots, n\}$, edges $\{e_1, \dots, e_m\}$.*

Let $Q = (q_{ij})$ with $q_{ij} = \begin{cases} -1 & \{i, j\} \in E \\ 0 & i \neq j : \{i, j\} \notin E \\ \deg(i) & i = j \end{cases}$ Let Q_{ij} be the matrix obtained from Q by removing the i -th row and the j -th column. Then the number of labelled spanning trees of G is $|\det(Q_{ij})|$, for any choice of i, j .

Corollary 2.7 (Cayley's formula). *The number of labelled spanning trees of K_n is n^{n-2} if $n \geq 2$.*

Proof. Define Q as in (2.5).

$$Q_{11} = \begin{pmatrix} n-1 & 1 & \cdots & 1 \\ 1 & \ddots & & \vdots \\ \vdots & & \ddots & 1 \\ 1 & \cdots & 1 & n-1 \end{pmatrix} \rightarrow \bar{Q}_{11} = \begin{pmatrix} n-1 & -1 & -1 & \cdots & -1 \\ -n & n & 0 & \cdots & 0 \\ -n & 0 & \ddots & & \vdots \\ \vdots & \vdots & & \ddots & 0 \\ -n & 0 & \cdots & 0 & n \end{pmatrix} \rightarrow \hat{Q}_{11} =$$

$$\begin{pmatrix} 1 & \star & \cdots & \star \\ 0 & n & & \vdots \\ \vdots & & \ddots & \star \\ 0 & \cdots & 0 & n \end{pmatrix}$$

$$\implies |\det(Q_{11})| = |\det(\hat{Q}_{11})| = n^{n-1-1} = n^{n-2} \quad \square$$

3 Planar Graphs

Definition 3.1. A **planar drawing** of a graph G consists of injective continuous functions

$$\phi_v : V(G) \rightarrow \mathbb{R}^2 \qquad \phi_e : [0, 1] \rightarrow \mathbb{R}^2$$

s.t.:

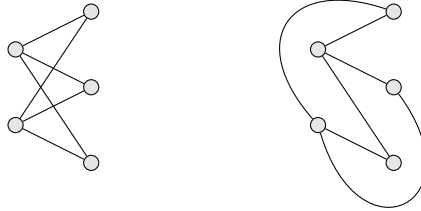
- $\{\phi_{\{u,v\}}(0), \phi_{\{u,v\}}(1)\} = \{\phi_u(u), \phi_v(v)\}$ for all $\{u, v\} \in E(G)$
- The sets $\phi_e((0, 1))$ for each $e \in E(G)$ are pairwise disjoint and disjoint from $\text{Im}(\phi_v)$ (i.e. no crossings)

A graph G is called **planar** if it has a planar drawing.

A **plane graph** is a planar graph, together with a fixed planar drawing.

Example 3.2.

- Trees
- K_4
- $K_{2,3}$



Definition 3.3. A **face** of a plane graph G is a maximal connected part of $\mathbb{R}^2 \setminus (\text{Im}(\phi_v(u)) \cup \text{Im}(\phi_e))$. Every plane graph has exactly one unbounded face called the **outer face**. All other faces are called **inner faces**. The **boundary** of a face are the vertices and edges of G incident with it. A plane graph is called **planar triangulation** if every face is bounded by a cycle of length precisely 3 (this includes the outer face).

Theorem 3.4 (Euler's formula). *Let $G = (V, E)$ be a connected planar graph. Then the number of faces $|F|$ of G is $|F| = |E| - |V| + 2$.*

Proof. Induction on $|E|$.

Since G is connected, we have $|E| \geq |V| - 1$ so we start the induction at $|E| = |V| - 1 \implies G$ is a tree. Since a tree does not contain a cycle, we have $|F| = |E| - |V| + 2$. Now let $|E| > |V| - 1$. Then G is not a tree, so G contains a cycle C . Choose an edge e on C and remove it to obtain a plane graph G' . e separates two faces in G , which fuse to a single face in G' . By induction, the number of faces $|F'|$ in G' is $|F| - 1 = |F'| = |E| - 1 - |V| + 2 \implies |F| = |E| - |V| + 2$. $\square$

Lemma 3.5. *For every plane graph $G = (V, E)$ with $|V| \geq 3$, we have $|E| \leq 3|V| - 6$.*

Proof. Wlog G is connected. Consider a planar drawing of G and let F be the set of faces. Every face is bounded by at least 3 edges and each edge bounds at most 2 faces. By Euler's formula (3.4), we obtain $\frac{2}{3}|E| \geq |F| = |E| - |V| + 2 \implies |E| \leq 3|V| - 6$. $\square$

Remark 3.6. This means that the number of edges in a planar graph grows only linear proportional to the number of vertices.

Example 3.7.

- K_5 is not planar (10 edges) due to (3.4)
- $K_{3,3}$ is not planar (without proof)

Definition 3.8. A **subdivision** of a graph G is a graph obtained from G by repeatedly subdividing some of its edges.

Theorem 3.9 (Kuratowski's theorem). *A graph G is planar iff G contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.*

Remark 3.10. A subdivision of a graph is planar iff the graph is planar

Definition 3.11. Let G be a graph. The graph obtained from G by contracting $e = \{u, v\} \in E(G)$ denoted G/e , is the graph obtained from G by deleting u, v , inserting a new vertex w and making w adjacent to all vertices in $N_G(u) \cup N_G(v)$. A graph G' is a **minor** of G if G' can be obtained from G by a sequence of vertex deletions, edge deletions and edge contractions.

Remark 3.12. If G' is a minor of G then G' planar $\iff G$ is planar.

Theorem 3.13 (Wagner's theorem). *A graph G is planar iff $K_{3,3}$ and K_5 are not a minor of G .*

4 Flows, Cuts, Connectivity

Definition 4.1. A **directed graph** is a pair (V, E) with $E \subseteq (V \times V) \setminus \{(x, x) : x \in V\}$. Write $\delta_G^+(v)$ for edges starting in v and $\delta_G^-(v)$ for edges ending in v .

Definition 4.2. A **flow network** is a tuple (G, s, t, c) where G is a graph, $s, t \in V(G)$, $s \neq t$ and $c : E \rightarrow \mathbb{R}_0^+$.

- s is called a **source**
- t is called a **target**
- $\forall e \in E(G) : c(e)$ is the **capacity** of e

A **flow** in a flow network is a function $f : E \rightarrow \mathbb{R}_0^+$ s.t.

- $f(e) \leq c(e)$ for all $e \in E(G)$
- $\forall v \notin \{s, t\}, \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$ (i.e. what flows in flows out)
- $\forall e \in \delta^-(s) \cup \delta^+(t) : f(e) = 0$ (i.e. nothing flows into a source or out of a target)

The **value** of a flow f is $\text{val}(f) = \sum_{e \in \delta^-(v)} f(e) = \sum_{e \in \delta^+(v)} f(e)$

Remark 4.3. For $v \in V$, write $f(\delta^+(v)) = \sum_{e \in \delta^+(v)} f(e)$; $f(\delta^-(v)) = \sum_{e \in \delta^-(v)} f(e)$

Definition 4.4. A **cut** in a graph/directed graph G is a partition of $V(G)$ into two non-empty parts A and B (i.e. $V(G) = A \dot{\cup} B$ and $A, B \neq \emptyset$). For $s, t \in V(G)$, $s \neq t \implies (A, B)$ is a (s, t) -cut if $s \in A, t \in B$.

If (S, T) is a (s, t) -cut, write $\delta^+(S) = \{e \in E(G) : e \text{ start in } S \text{ ends in } T\} = \delta^-(T)$ and $\delta^-(S) = \{e \in E(G) : e \text{ starts in } T \text{ and ends in } S\} = \delta^+(T)$.

The **capacity** of the cut is $c(S, T) = \sum_{e \in \delta^+(S)} c(e) = c(\delta^+(S)) = c(\delta^-(T))$.

Theorem 4.5 (max flow-min cut theorem). *Let (G, s, t, c) be a flow network. Then $\max\{\text{val}(f) : f \text{ flow}\} = \min\{c(S, T) : (S, T) \text{ is a } (s, t)\text{-cut}\}$.*

Lemma 4.6. *Let f be a flow, (S, T) be a (s, t) -cut. Then $\text{val}(f) = f(\delta^+(S)) - f(\delta^-(S))$.*

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